Assignment 0

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1 Introduction

The goal of this assignment is to analyze a real-world network and study its main structural properties. The selected network is built from anime data collected from MyAnimeList. The network represents similarity relationships between anime based on their genre composition.
Each node represents an anime, while edges represent similarity relationships between anime that share common genres.

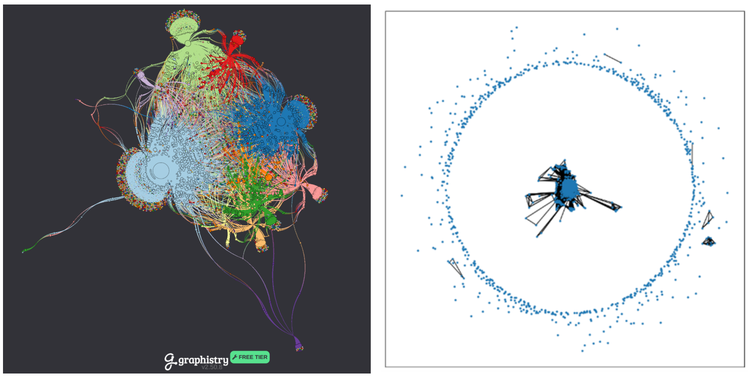

Figure 1: Left: Graph visualized dynamically on the Graphistry website. Right: NetworkX graph.

2 Dataset

The original dataset contains 29816 anime described by 39 attributes, including metadata such as genres, themes, scores, and popularity indicators.
To ensure data quality and relevance, the dataset was filtered according to the following criteria:

- at least 50000 members on MyAnimeList, to ensure statistical relevance;
- removal of entries without genre information;
- removal of duplicate anime using the MAL ID attribute.

After preprocessing, the final dataset contains 3579 anime, which represent the nodes of the network. Only the genres attribute was used for graph construction because it provides a consistent and interpretable criterion for defining similarity between anime. Other attributes were excluded because they were incomplete or not directly related to the structural analysis of the network.

3 Network Construction

The network is modeled as an undirected graph and the construction rules are:

- nodes represent anime;
- an edge is created between two anime if they share at least two genres.

The threshold of two shared genres was chosen to reduce weak connections caused by very common genres such as “Action” or “Comedy”, creating a network where edges represent more significant similarities. The final graph contains 3579 nodes and 780393 edges.

4 Network Characterization

Several network metrics were computed to analyze the topology of the graph. The degree distribution was analyzed using a power-law fitting approach.

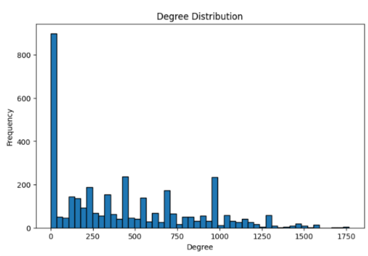

Figure 2: Degree distribution.

The degree distribution was tested against a power-law model, yielding $\alpha = 2.897$ and $x_{\min} = 454$. Here, α controls the decay of the degree distribution, while $x_{\min}$ indicates the minimum degree from which the power-law fit is considered.
Here, R and the p-value are obtained from the `distribution_compare()` function of the `powerlaw` library. R indicates which model is preferred ($R < 0$ favors the alternative model), while the p-value indicates whether this difference is statistically significant ($p < 0.05$).

Comparison	R	p	Best model
Power law vs Exponential	-187.794	< 0.001	Exponential
Power law vs Lognormal	-196.488	< 0.001	Lognormal
Power law vs Truncated Power Law	-186.996	< 0.001	Truncated Power Law
Power law vs Stretched Exponential	-246.137	< 0.001	Stretched Exponential

Since all comparisons result in $R < 0$ and $p < 0.05$, the alternative distributions are statistically preferred over the pure power-law model.
Compared with an Erdős–Rényi random graph of the same size and density, the network has a similar average degree (436.10 vs. 436.67) and average shortest path length (1.932 vs. 1.878), but a much higher clustering coefficient (0.6296 vs. 0.1221). Therefore, the network is not purely random and does not follow a strict power-law distribution. Its combination of high clustering and short paths suggests small-world-like properties, similar to those observed in Watts–Strogatz networks.

5 Network Analysis

To identify the most important nodes, Degree Centrality was computed. This metric measures how many direct connections each node has compared to the maximum possible number of connections.

Node	Degree Centrality
Zero no Tsukaima	0.493851
Kurenai no Buta	0.477921
Groove Adventure Rave	0.474008
One Piece Film: Red	0.438513
Mahou Sensei Negima!	0.436277

These anime act as hubs because they share common genres with many other anime in the network, resulting in a large number of connections.
The shortest path analysis was performed on the largest connected component and returned a diameter of 5 and an average shortest path length of 1.932. Despite the size of the network (3579 nodes), nodes can be reached through a small number of steps.
The network density is 0.121882, with an average degree of 436.10, indicating a relatively dense structure. This is mainly due to the graph construction rule, where anime sharing at least two genres are connected, naturally generating many links between similar nodes.
The average clustering coefficient is 0.6296, showing that nodes tend to form tightly connected groups. The assortativity coefficient is 0.1745, indicating a weakly positive correlation between node degrees, meaning that nodes with similar connectivity tend to connect more frequently.

6 Conclusion

The anime similarity network exhibits a structured complex-network topology rather than a purely random or strict scale-free structure. Its degree distribution is better described by a stretched exponential than by a pure power law. The network shows high clustering, short paths, moderate positive assortativity, and highly connected nodes.
Compared with an Erdős–Rényi network of similar size and density, it has substantially higher clustering while maintaining similarly short paths. Overall, these properties suggest a small-world-like topology driven by genre-based similarity between anime.